

A AegisEPC Documentation

Detailed Structured Engineering & Technical Platform Manual

Repo: <https://github.com/Jaswanth258/AI-Data-Centre-EPC-Platform>

1. System Overview & Core Concept

AegisEPC is an enterprise-grade AI-powered project intelligence layer designed to mitigate procurement misalignment and quality compliance bottlenecks in Engineering, Procurement, and Construction (EPC) projects (such as hyperscale data centre builds).

The platform replaces traditional Vector DB-based RAG engines with **Clause-Tree Retrieval**, which structures engineering documents into deterministic section trees to eliminate chunk truncation errors and guarantee 100% citation traceability.

2. Architecture & Agent Workflows

2.1 Agent 1: RFI & Knowledge Copilot

Allows engineers to query specifications in natural language. The agent retrieves the relevant clause index via table-of-contents navigation, fetches exact text bodies, and answers queries with strict citations. Supports per-document memory isolation.

2.2 Agent 2: Quality Compliance Agent

Automates vendor submittal reviews. Uploading a submittal extracts all engineering parameters, matches them dynamically to spec clauses, computes verdicts (Compliant/Deviation), flags warning levels, and exports CSV logs.

3. Clause-Tree vs Vector RAG

FEATURE	TRADITIONAL VECTOR RAG	AEGISEPC CLAUSE-TREE RAG
Ingestion Method	Splits text arbitrarily every 500 tokens	Parses document headers into structured nodes
Context Retention	Breaks lists/clauses across chunk limits	Retains full clause text and section hierarchy
Search Mechanism	Vector cosine similarity	TOC navigation + exact dictionary lookup
Hallucination Risk	High (retrieves irrelevant semantic text)	Zero (retrieves exact specified clause text)

4. Ingestion & Document Parsing

The parser (defined in `backend/doc\_parser.py`) indexes markdown headings (`#`, `##`, `###`) into distinct section-clause maps. Bullet points, parameter descriptions, and numerical requirements are grouped inside their respective parent nodes. The PDF parser uses `pdfplumber` to extract clean text while retaining layout structure.

5. Installation & Execution

```
git clone https://github.com/Jaswanth258/AI-Data-Centre-EPC-Platform.git
cd AI-Data-Centre-EPC-Platform
python -m venv .venv
.venv\Scripts\activate
pip install -r backend/requirements.txt
$env:GEMINI_API_KEY="your-gemini-key"
python -m uvicorn backend.main:app --host 127.0.0.1 --port 8001
```